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|-------------------------------------------------------|------------------------------------|
| Program: <b>Diploma in Food Processing Technology</b> |                                    |
| Course Code: <b>3423</b>                              | Course Title: <b>Food Analysis</b> |
| Semester: <b>3</b>                                    | Credits: <b>3</b>                  |
| Course Category: <b>Program core</b>                  |                                    |
| Periods per week: <b>3 (L:2 T:1 P: 0)</b>             | Periods per semester: <b>45</b>    |

### Course Objectives:

1. To learn about government regulations, regulating bodies and accreditation bodies.
2. To emphasis on various methods of food component analysis
3. To impart knowledge about various chromatographic separation techniques.

### Course Prerequisites:

| Topic                                                | Program/Course Name                    |
|------------------------------------------------------|----------------------------------------|
| <b>Basic knowledge of food processing technology</b> | Fundamental food processing technology |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                    | Duration (Hours) | Cognitive Level |
|-----------------|--------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Describe the government regulations pertaining to food analysis                | 5                | Understanding   |
| CO2             | Review the principles behind the analysis of various food components           | 15               | Understanding   |
| CO3             | Recognize the different techniques employed for analysis of components in food | 12               | Understanding   |
| CO4             | Explain the instrumentation and working of equipment used for food analysis    | 13               | Understanding   |

## CO–PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 2   |     |     |     |     |     |     |
| CO2             | 2   |     |     |     |     |     |     |
| CO3             | 2   |     |     |     |     |     |     |
| CO4             | 2   |     |     |     |     |     |     |

3-Stronglymapped, 2-Moderatelymapped, 1-Weaklymapped

## Course Outline:

| Module Outcomes                                                                                                                                                                                    | Description                                                                           | Duration (Hours) | Cognitive Level |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>                                                                                                                                                                                         | <b>Describe the government regulations pertaining to food analysis</b>                |                  |                 |
| M1.01                                                                                                                                                                                              | List the significance and types of analysis                                           | 1                | Understanding   |
| M1.02                                                                                                                                                                                              | Describe the regulations of FSSAI 2006, NABL accreditation                            | 2                | Understanding   |
| M1.03                                                                                                                                                                                              | Discuss CAC, AOAC-Export Inspection Council                                           | 2                | Understanding   |
| <b>Contents:</b> Significance and types of analysis - Government regulations - regulations of FSSAI 2006- NABL accreditation -CAC -AOAC - Export Inspection Council                                |                                                                                       |                  |                 |
| <b>CO2</b>                                                                                                                                                                                         | <b>Review the principles behind the analysis of various food components</b>           |                  |                 |
| M2.01                                                                                                                                                                                              | Explain proximate analysis                                                            | 4                | Understanding   |
| M2.02                                                                                                                                                                                              | Review the analysis of extraneous matter                                              | 5                | Understanding   |
| M2.03                                                                                                                                                                                              | Describe thermal Analysis                                                             | 5                | Understanding   |
|                                                                                                                                                                                                    | Series Test –I                                                                        | 1                |                 |
| <b>Contents:</b> Proximate analysis – Moisture and Total Solids - Ash, Fat, Carbohydrate, Protein Analysis, pH and Titratable Acidity, Analysis of extraneous matter - Thermal Analysis – TGA, DSC |                                                                                       |                  |                 |
| <b>CO3</b>                                                                                                                                                                                         | <b>Recognize the different techniques employed for analysis of components in food</b> |                  |                 |

|                                                                                                                                                                     |                                                                                    |   |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---|---------------|
| M3.01                                                                                                                                                               | Define the concept of Spectroscopy, its principles                                 | 4 | Understanding |
| M3.02                                                                                                                                                               | State beer lamberts law and deviation from beer lamberts law                       | 4 | Understanding |
| M3.03                                                                                                                                                               | Summarize UV Visible spectroscopy, Fluorescence spectroscopy                       | 4 | Understanding |
| <b>Contents:</b> Spectroscopy principles - Beer Lambert's law - deviation from Beer Lambert's law- UV Visible spectroscopy, Fluorescence spectroscopy               |                                                                                    |   |               |
| <b>CO4</b>                                                                                                                                                          | <b>Explain the instrumentation and working of equipment used for food analysis</b> |   |               |
| M4.01                                                                                                                                                               | Outline the principles of Chromatography                                           | 4 | Understanding |
| M4.02                                                                                                                                                               | Recognize different chromatographic techniques                                     | 4 | Understanding |
| M4.03                                                                                                                                                               | Describe High performance liquid chromatography                                    | 4 | Understanding |
|                                                                                                                                                                     | Series Test–II                                                                     | 1 |               |
| <b>contents:</b> Principles of Chromatography- Column chromatography - Thin layer chromatography - Gas-liquid chromatography-High performance liquid chromatography |                                                                                    |   |               |

### Text / Reference

| T/R | Book Title/Author                                                                                                                               |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|
| R1  | Nielsen, Suzanne (Ed.), Food Analysis, 2010, Springer US, ISBN 978-1-4614-2589-2                                                                |
| R2  | Semih Otles, "Methods of Analysis of Food Components and Additives", Published October 17, 2016, by CRC Press, ISBN 9781138199149               |
| R3  | Semih Otles, Handbook of Food Analysis Instruments, 2016, CRC Press, ISBN 9780429147340                                                         |
| R4  | Adriana S. Franca and Leo M.L. Nollet, Spectroscopic Methods in Food Analysis, 2018, CRC Press, ISBN 9781498754613 (978-1-4987-5461-3)          |
| R5  | Leo M. L. Nollet, Handbook of Food Analysis Vol I and II, Second Edition, Revised and Expanded, 2004 by Marcel Dekker, Inc, ISBN: 0-8247-5036-5 |

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| R6 | Rui M. S. Cruz, Igor Khmelinskii, Margarida Vieira, Methods in Food Analysis, 2014, CRC Press, ISBN 9781138582477 |
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### Online Resources

| Sl.No | WebsiteLink                                                                                                                                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <a href="https://www.vedantu.com/physics/beer-lambert-law">https://www.vedantu.com/physics/beer-lambert-law</a>                                                                                                               |
| 2     | <a href="https://microbenotes.com/thin-layer-chromatography/">https://microbenotes.com/thin-layer-chromatography/</a>                                                                                                         |
| 3     | <a href="https://www.chemistryviews.org/details/education/9464911/What_is_HPLC/">https://www.chemistryviews.org/details/education/9464911/What_is_HPLC/</a>                                                                   |
| 4     | <a href="https://www.microlit.com/what-are-the-guidelines-of-nabl-accreditation-process-and-why-is-it-important/">https://www.microlit.com/what-are-the-guidelines-of-nabl-accreditation-process-and-why-is-it-important/</a> |
| 5     | <a href="https://www.britannica.com/science/chromatography">https://www.britannica.com/science/chromatography</a>                                                                                                             |